

DP024

Physics 2

Semester 2

Session 2022/2023

1 hour 45 minutes



**KOLEJ
MATRIKULASI
MELAKA**

PROGRAM DRAW A QUESTION (DAQ)

MODEL PAPER SEMESTER 2

SET A

PHYSICS UNIT

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

INSTRUCTIONS TO CANDIDATE:

The question paper consists of 6 questions.

Answer **all** questions.

The use of electronic calculator is permitted.

QUESTION	MARKS
1	/16
2	/12
3	/10
4	/14
5	/14
6	/14
TOTAL	/80

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deutron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|---|---|
| 1. $x = A \sin \omega t$ | 18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$ |
| 2. $\omega = \frac{2\pi}{T} = 2\pi f$ | 19. $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$ |
| 3. $v = A\omega \cos \omega t$ | 20. $\tau = RC$ |
| 4. $a = -A\omega^2 \sin \omega t$ | 21. $Q = Q_o e^{\frac{-t}{RC}}$ |
| 5. $k = \frac{2\pi}{\lambda}$ | 22. $Q = Q_o \left(1 - e^{\frac{-t}{RC}} \right)$ |
| 6. $v = f\lambda$ | 23. $B = \frac{\mu_o I}{2\pi r}$ |
| 7. $y(x, t) = A \sin(\omega t \pm kx)$ | 24. $B = \frac{\mu_o NI}{L}$ |
| 8. $v = \frac{dy}{dt}$ | 25. $B = \frac{\mu_o I}{2r}$ |
| 9. $y = 2A \cos kx \sin \omega t$ | 26. $\phi = BA \cos \theta$ |
| 10. $v = \sqrt{\frac{T}{\mu}}$ | 27. $\Phi = N\phi$ |
| 11. $I = \frac{dQ}{dt}$ | 28. $\varepsilon = -\frac{d\phi}{dt}$ |
| 12. $Q = ne$ | 29. $\varepsilon = Blv \sin \theta$ |
| 13. $\rho = \frac{RA}{l}$ | 30. $\varepsilon = -NA \frac{dB}{dt}$ |
| 14. $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$ | 31. $\varepsilon = -NB \frac{dA}{dt}$ |
| 15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ | 32. $R = 2f$ |
| 16. $V = IR$ | |
| 17. $C = \frac{Q}{V}$ | |

$$33. \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$34. \quad m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$35. \quad y_m = \frac{m\lambda D}{d}$$

$$36. \quad y_m = \frac{(m + \frac{1}{2})\lambda D}{d}$$

$$37. \quad \Delta y = \frac{\lambda D}{d}$$

1. a) A progressive wave equation is given below.

$$y = 6 \sin (16 \pi t + 0.8 x)$$

where y and x in cm, and t in s. Determine

- the velocity of particle at position $x = 4 \text{ cm}$, and $t = 0.1 \text{ s}$,
- velocity of the wave.

[6 marks]

- b) Two harmonic waves are represented by the equations:

$$y_1 = 6 \sin (2\pi t + 2\pi x)$$

$$y_2 = 6 \sin (2\pi t - 2\pi x)$$

where y_1 , y_2 and x are in meters, and t in seconds.

- Determine the amplitude of the new wave, angular frequency and wave number.
- Write an expression for the new wave when both waves are superimposed.

[5 marks]

- c)

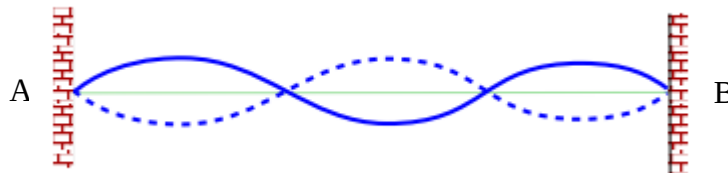


FIGURE 1

A nylon guitar string has a mass per unit length, μ of 14.4 g m^{-1} and is under a tension of 300 N . The fixed supports (A and B) are 180 cm apart. The string is oscillating in the standing wave pattern shown in **FIGURE 1**. Calculate

- the speed, v ,
- wavelength, λ .

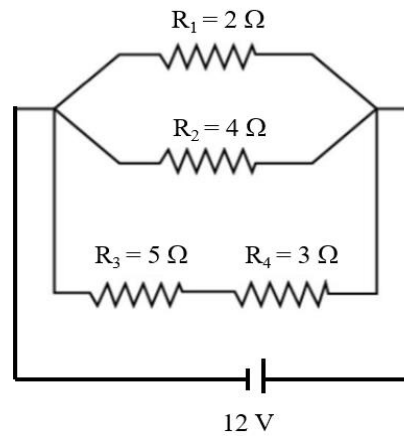
[5 marks]

2. a) The number of electrons, $n = 1.78 \times 10^{23}$ passes through a light bulb in 1 hour. Calculate the magnitude of current flows in the bulb.

[2 marks]

- b) A copper wire with a cross-sectional area of 60.0 mm^2 has a resistance of 0.50Ω . The resistivity of copper wire is $1.68 \times 10^{-8} \Omega \text{ m}$. Calculate the length of wire.
- [2 marks]

c)

**FIGURE 2**

For the circuit in **FIGURE 2**, calculate

- i) the effective resistance of the circuit,
- ii) the total current,
- iii) the voltage across the 4Ω resistor.

[8 marks]

3. a)

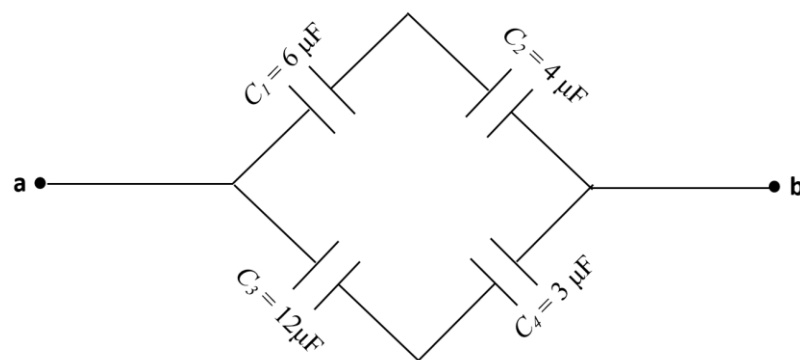
**FIGURE 3**

FIGURE 3 shows four capacitors arranged between point **a** and **b**. Calculate the

- i) effective capacitance,
- ii) charge on C_3 when 12 V applied between point **a** and **b**.

[6 marks]

b)

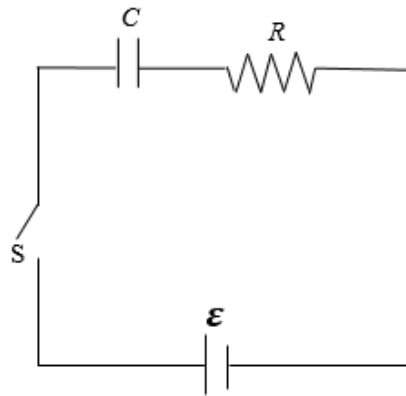
**FIGURE 4**

FIGURE 4 consists of an uncharged capacitor, C . Sketch a graph of

- i) current against time
- ii) charge against time

when switch S is closed and label the value of charge and current at time constant.

[4 marks]

4. a) A 4.0 m long wire is made into a coil of 10 turns. If the current in the wire is 20 mA,
- i) calculate the magnetic field at the centre of the coil.
 - ii) show the direction of the magnetic field for a coil by using a simple sketch.

[6 marks]

b)

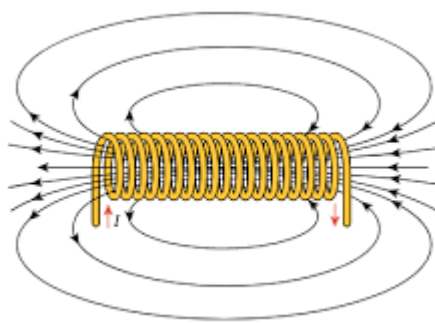
**FIGURE 5**

FIGURE 5 shows 1200 turns of solenoid.

- i) Determine the current that flows in the solenoid of length 1.5 m to produce a magnetic field of 2.0 mT at the centre.

- ii) Determine the magnetic field at the centre if the length of the solenoid is changed to 1.8 m.
[2 marks]
- c) A solenoid has 150 turns and a cross sectional area of 5 cm^2 . If the solenoid is transferred from a region without magnetic field to another region with 0.6 T in 0.05 s, find the change in magnetic flux and the induced emf in the solenoid.
[4 marks]
5. a) A dentist uses a small dental mirror to check patients' teeth. When it is placed 2.5 cm from a molar tooth, the mirror will provide an image that is twice the size as the tooth.
- What is the image distance of the tooth?
 - Is the image real or virtual?
 - Determine the radius of curvature for the mirror.
 - Sketch a labelled ray diagram to show the formation of the tooth's image.
- [7 marks]
- b) A diverging lens has a focal length of 20.0 cm forms an erect and virtual image where the position of image is $\frac{1}{3}$ of the position of the object.
- Where should the object be placed?
 - Sketch a labelled ray diagram to show the formation of the image.
- [7 marks]
6. a) The separation of a double-slit is 0.40 mm. A beam of a monochromatic light of wavelength 650 nm is incident on this double slit. If the screen is located 1.3 m away from the slits, find the distance from the central bright fringe to the
- third bright fringe,
 - second dark fringe.

[6 marks]

- b) The slit separation in a Young's double-slit experiment is found to be 0.6 mm. The distance of the screen from the slits is 1.5 m and the wavelength used is 3.0×10^{-7} m.
- Calculate the fringe separation.
 - How will the fringe pattern be affected if the distance between the slits and the screen is increased?

[4 marks]

- c) A monochromatic light with wavelength 550 nm passes through a double slit. If the slit separation is 0.5 mm and a screen is placed 1.2 m in front of the slits, determine the distance from the 3rd dark fringe to the 9th bright fringe from the same side.

[4 marks]

- END OF QUESTION PAPER -